

	Company / Factory / Centre / Area :	Document Reference :	Revision number :
	AECI MINING EXPLOSIVES	RS/ZIM/SHEQ/0010	02
	Document Type : RECORD SHEET		TCF Number -
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MMU:	SITE:	DATE:
CONTROLLER:	BLAST:	PRODUCT:
No. blastholes:	Ave. Depth (m) : Diameter (mm) :	B x S (m) : Stemming (m) :

Dry holes.....Wet holesDe-watered.....Bottom loaded.....Top loaded..... Augured.....

Previous Audit (P):

Rating: 0 = No; 1 = Yes

QUALITY CHECK	NOTES	P	0/1	SHE CHECK	P	0/1
1. Pre-start & DILO correct and signed off.				1. PPE correct (hardhat, safety boots, glasses, gloves & reflective vest)?		
2. MMU Presentation				2. Safety belts in use?		
3. Spring-scale ID, zeroed & condition: Clear Sample Cups				3. Hearing protection, dust masks, safety harness available if needed		
4. 500g checkwt, ID. or water 566g ±				4. 1 st Aid kit (complete), eye wash and drinking water available?		
5. Sampling intervals : initial (after 1 st hole & 20 min or every 4 tons)				5. Hazchem stickers, Tremcard + Manu. Permit, Class II, licence & CTP - Valid?		
6. Cup density requirement specified and approved (by customer)				6. Operator passed out on MMU, PDP and Drivers Licence, induction and mine licence valid?		
7. End of Hose Cup Density in spec. Hopper Density in spec.	g/cc g/cc			7. Min. waste on start up & shut down. Any process waste during charging?		
8. Gassing rate correct?				8. Reverse hooter operational?		
9. ANFO Oil content	%			9. Guards and inspection points fit and secure.		
10. Product appearance & Temp.	°C			10. Confined space stickers in place.		
11. Product level in hopper correct -No air or overflow				11. Safety rails in use.		
12. MMU Quality Sheet completed on bench & signed. (SA1501/SA1502)				12. Tyres good (Incl. Spare) and tools? Wheel chokes being used (x2)		
13. Operating procedures in place and followed?				13. Fire Extinguisher secured, pressured and within inspection date?		
14. Hydraulic oil temperature (40 – 60 °C)	°C			14. Any loose articles not secured inside and outside cab?		
15. MMU last calibrated (3 monthly) 20kg ±1kg(<200kg/m)				15. Hose handling : good use of legs & PPE and faces edge if within 3m ?		
16. Pumping rate OK? Hose lube correct (max.2%)	%			16. No leaks. (Diesel, hydraulic oil, FO, water and gassing)		
17. Instruments correct–gassing flow-meter & FO totaliser.				17. Safe driving – use spotter to reverse to avoid driving over primed holes.		
18. Primer size and located correctly: no floaters and tippet used.				18. Safety awareness - use 3 limb contact in/out of cab.		
19. Hose down to bottom of blast hole and slowly withdrawn.				19. Pump trip correct and working with pump inspections up to date?		
20. Stemming measured and correct? Time allowed for gassing?				20. Bursting disc(s) correct rating? Press Gauge functional (PP 30Bar-Red, EP 20Bar-Yellow)		
				21. 5 Steps to Safety?		
% QUALITY SCORE* = no. of points x 100 / no. of applicable items =				% SHE SCORE* = no. of points x 100 / no. of applicable items =		

* no award 0-16.6% Half bronze 16.7-33.3% Bronze 33.3-49.9% Half silver 50-66.6% Silver 66.7-83.2% Half gold 83.3-99.9% Gold 100%

DONE BY:

DATE:

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ON BENCH MMU INSPECTION

QUALITY REQUIREMENTS:

- Pre-start inspection & DILO** completed, signed off and agrees with actual condition?
- MMU Presentation** looks professional with inside/outside of cab clean, neat signs/logos and all items mounted correctly with no leaks.
- Springscale** uniquely identified, in good condition and zeroed. Only clear cup samples to be used.
- Springscale** checked daily with 500g check weight and/or water 566g cup and recorded on MMU Quality sheet.
- Minimum **sampling** frequency for cup densities complied after 1st hole, thereafter every 20min or every 4 tons there after. (Customer may request different frequency)
- Cup density** range specified in MMU file or covered by approved Change Proposal. Approved by Customer.
- Gassing rate** of product measured and within range specified.
- Product Cup densities at end of hose** and (if possible) in **hopper** checked, recorded and within spec.
- If loading **ANFO**, **Oil** content calculated if MMU fitted with oil totaliser and if suspect, sample taken to test oil content.
- Product appearance** is off-white paste containing AN Porous Prill for blends, brownish for P100 and ANFO to have light pink colour indicating the correct addition of fuel oil. Check with blaster if he is happy with the product.
- Correct **hopper levels** with no trapped air or overflow.
- MMU Quality sheet** filled in correctly before leaving bench with no blanks left and signed by blaster. (SA1501/2)
- Operating instructions** available in MMU Manual and being followed especially as relating to quality and SHE. A PJO can also be done.
- Hydraulic oil** in temperature range with cooling system working correctly and no leaks in system.
- Within **Calibration Date** (Every 3 months). Pumping rate < 200kg/min - check correct charge per hole by measuring **MMU output** using 20kg scale & bucket, ensuring hopper and hose are full of product before loading 20kg of product into the pre-weighed bucket.
- Correct pumping rate being used and check the **hose lube** flowrate is not more than 2% of MMU output rate.
- All **instruments** such as meters and gauges are readable and in working order especially gassing flowmeter and oil totaliser.
- Correct **Primers** used and located properly in blasthole with no "floaters". A Booster Tippet fits onto the end of the hose and holds the booster in place when loading wet holes. (hole diam. <100mm – 150g booster, >100mm – 400g booster)
- Hose** placed to bottom of hole and then pulled up slowly while pumping.
- Are **stemming** heights correct and product given sufficient time to gas fully before stemming.

SHE REQUIREMENTS:

- Are hard hat, gloves, safety boots safety glasses, reflective vests being worn and are hearing protection and dust mask available.
- Are **safety belts** fitted into the MMU and being used?
- Excessive noise requires hearing protection, dusty conditions dust masks and working at heights the correct safety harness.
- First aid** (complete kit), eye wash bottle (vacu-pack) and drinking water available in MMU.
- Hazchem** signs are required on sides of MMU indicating the product being transported. Clearly visible and in good order? Is vehicle **licence**, **Tremcard** and both **Manufacturing Permits** (a. Permission to Manufacture Blasting Materials from CIE, b. Schedule II of Explosives Workplace Licence from Dept. of Labour in the MMU) up to date? (AN-1942; EP-3375; Gassing Solution-3139; FO & diesel (>500l) –1202. Multi-load sign in front and back.
- Is operator passed out on relevant MMU, valid licence, PDP, mine licence and Induction for relevant mine?
- How much **waste** is produced on start-up and shut-down, any process waste on bench and what is done with it?
- MMU fitted with **reverse hooter** that is operational.
- Are all **guards and inspection points** fitted and secure?
- Confined space stickers** in place and on entry points to confined areas.
- Working at heights requires **safety rails** to be used or needs improving if only safety-harness available.
- Tyres** (incl. Spare tyre) are to have at least 1mm of tread across their surface and be correctly inflated at all times. A pressure gauge to be available at the site if not in MMU.
- Fire extinguisher(s)** with a total of 9kg are to have a double tie to prevent accidental release, indicate full pressure and be inspected annually.
- There are to be no **loose articles** on & in the MMU that are not properly secured or locked away.
- Hose handling** is done safely by minimising back strain by not bending, but using the legs, while using gloves to protect the hands and safety glasses. When working within 3m of a highwall always face the edge of the bench.
- No leaks** of diesel, hydraulic oil, FO, water and gassing allowed, except for short periods, when **drip trays** used until leak repaired.
- All Mine road rules are complied with to ensure **safe driving**. Primed holes and accessories are not driven over and a **spotter** is used to ensure safe reversing.
- Safety aware** at all times, especially on the blast, ie use 3 limb contact getting in and out of the cab and watch where you walk especially near the highwall.
- Pump trip** mechanism in place and working and inspections up-to-date (flow sensor or H/L pressure).
- Bursting disc(s)** correct for product pump (30bar-red) and emulsion pump (20bar-yellow). Spare bursting disc in cab.
- The **4 Steps to Safety Questions** 1. Is this a safe situation? 2. Is there any equipment, material, machinery or conditions which are unsafe? 3. Are there any of my colleagues doing anything unsafe? 4. What am I going to do about it now?
In addition, how far from the MMU or blast site may one go to have a cigarette? (10m)